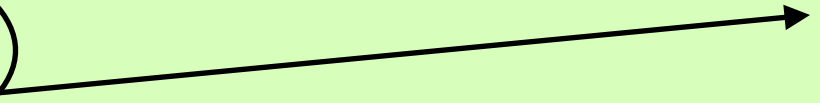


Welcome! 🍌

EECS 376, Sec. 18

D2: Divide and conquer

Sam Hiken



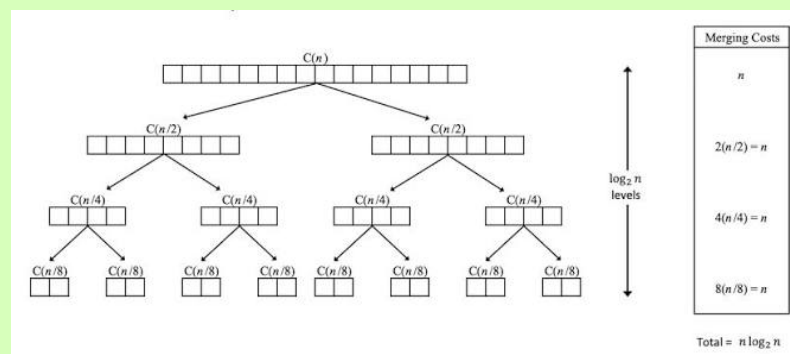
Admin note

I will have a substitute next week

Divide and conquer review

Divide and conquer algorithms have the following parts:

- **Base case:** small input, output the answer directly
- **Otherwise:**
 - Divide: split the input into smaller pieces, recursively call the algorithm on the smaller pieces
 - Conquer: combine the outputs of the recursive calls



Proving correctness

Usually you use proof by induction

1. Show the algorithm is correct on the base case
2. Show that
 1. If the output of the recursive calls is correct,
 2. Then the combined output from the conquer step is also correct

Proving runtime

Two steps:

1. **Derive a recurrence relation:**

1. Let $T(n)$ be the runtime on input of size n
2. Express $T(n)$ as a function of T on smaller inputs.

2. **Solve the recurrence relation**

Quiz

Suppose we have a divide and conquer algorithm where:

- Divide step splits input into 2 halves
- Conquer step takes time n^3

What would a recurrence relation look like?

A. $T(n) = T(n/2) + n^3$

B. $T(n) = 3T(n/3) + n^2$

C. $T(n) = 2T(n/2) + n^3$

Master Theorem

Formula for “solving” recurrence relations

- tells you the overall runtime using the recurrence relation

It appears on page 2 of your worksheet

Quiz: runtime of an algo with recurrence relation:

$$T(n) = 2T(n/2) + n^3$$

A. $T(n) = \Theta(n^3)$

B. $T(n) = \Theta(n^3 \log n)$

C. $T(n) = \Theta(n)$

Worksheet time!



<https://tinyurl.com/4w4jrbu8>